



GSSS | GEETHA SHISHU
SHIKSHANA SANGHA(R)
**INSTITUTE OF ENGINEERING
AND TECHNOLOGY FOR WOMEN**

(Affiliated to Visvesvaraya Technological University, Belagavi, Approved by AICTE New Delhi & Govt of Karnataka)
KRS Road, Metagalli, Mysuru - 570016



DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE



DATABASE MANAGEMENT SYSTEM (BCS403)

STUDENT NAME :

USN:

SEM AND SECTION:



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SHIKSHANA SANGHA(R)

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KRS ROAD, METAGALLI | MYSURU - 570 016 | KARNATAKA | INDIA



Phone : 0821 - 2581304,4257304 / 2977306 Fax : 0821 - 2581305 Email : gsssegg@gmail.com, principal@gsss.edu.in Web : www.geethashishu.in

DEPARTMENT OF ARTIFICIAL INTELLIGENCE & DATA SCIENCE

Laboratory Certificate

This is to certify that Miss
with USN..... has satisfactorily completed all
the experiments of
IPCC Laboratory with subject code of
Semester B.E Course during the Academic year 20 - 20

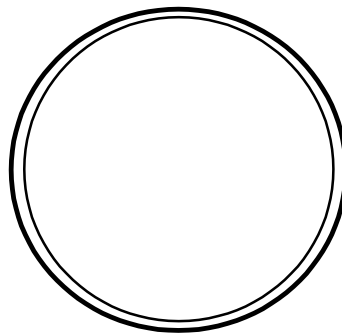
Student Signature

Faculty Signature

HoD Signature

Date:

Final Marks



In Words

.....

.....

Vision of the Department

To empower women engineers with cutting-edge knowledge and skills in Artificial Intelligence and Data Science, fostering innovation, leadership, and ethical practices to drive transformative changes in society and industry.

Mission of the Department

Mission 1: Provide a rigorous and comprehensive education in Artificial Intelligence and Data Science.

Mission 2: Foster a culture of research and innovation among students and faculty

Mission 3: Equip students with the skills and confidence to take on leadership roles in their careers

Mission 4: Promote ethical considerations and the responsible use of AI and Data Science to benefit society.

PROGRAM EDUCATIONAL OBJECTIVES

PEO 1: To demonstrate technical proficiency and a deep understanding of fundamental and advanced concepts in Artificial Intelligence and Data Science in solving complex engineering problems.

PEO 2: To engage in innovative practices and research, contributing to the development of new technologies and solutions by advancing the frontiers of knowledge and technology

PEO 3: To exhibit leadership qualities and the ability to work effectively in multidisciplinary teams, demonstrating strong communication, project management, and teamwork skills to drive successful project outcomes.

PEO 4: To uphold ethical standards and demonstrate a commitment to social responsibility, ensuring that their work positively impacts society and adheres to legal and ethical guidelines.

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SL No	Date	Name of the Experiment	Page No.	Marks				Signature of the faculty
				Conduction	Record	Viva	Total	

LAB LESSON PLAN – 2025-2026

Semester- IV

Subject Name: DATABASE MANAGEMENT SYSTEM [BCS403]

Faculty Name: Mamatha N

Course Outcomes (Course Skill Set) :

At the end of the course the student will be able to:

CO 1: Describe the basic elements of a relational database management system

CO 2: Design entity relationship for the given scenario.

CO 3: Apply various Structured Query Language (SQL) statements for database manipulation.

CO 4: Analyse various normalization forms for the given application.

CO 5: Develop database applications for the given real world problem.

CO 6: Understand the concepts related to NoSQL databases.

Program No	Programs	COs
1	Create a table called Employee & execute the following. Employee(EMPNO,ENAME,JOB,MANAGER_NO,SAL,COMMISSION) 1. Create a user and grant all permissions to the user. 2. Insert the any three records in the employee table contains attributes EMPNO,ENAME JOB, MANAGER_NO, SAL, COMMISSION and use rollback.Check the result. 3. Add primary key constraint and not null constraint to the employee table. 4. Insert null values to the employee table and verify the result.	CO1, CO3, CO4

2	Create a table called Employee that contain attributes EMPNO,ENAME,JOB, MGR,SAL & execute the following. 1. Add a column commission with domain to the Employee table. 2. Insert any five records into the table. 3. Update the column details of job 4. Rename the column of Employ table using alter command. 5. Delete the employee whose Empno is 105.	CO1, CO3, CO4
---	---	------------------------------

3	Queries using aggregate functions (COUNT, AVG, MIN, MAX, SUM), Group by, Orderby. Employee(E_id, E_name, Age, Salary) 1. Create Employee table containing all Records E_id, E_name, Age, Salary. 2. Count number of employee names from employee table 3. Find the Maximum age from employee table. 4. Find the Minimum age from employee table. 5. Find salaries of employee in Ascending Order. 6. Find grouped salaries of employees.	CO1, CO3, CO4
4	Create a row level trigger for the customers table that would fire for INSERT or UPDATE or DELETE operations performed on the CUSTOMERS table. This trigger will display the salary difference between the old & new Salary. CUSTOMERS(ID,NAME,AGE,ADDRESS,SALARY)	CO1, CO3, CO4
5	Create cursor for Employee table & extract the values from the table. Declare the variables, Open the cursor & extract the values from the cursor. Close the cursor. Employee(E_id, E_name, Age, Salary)	CO1, CO3, CO4
6	Write a PL/SQL block of code using parameterized Cursor that will merge the data available in the newly created table N_RollCall with the data available in the table O_RollCall. If the data in the first table already exist in the second table then that data should be skipped.	CO1, CO3, CO4
7	Install an Open Source NoSQL Data base MangoDB& perform basic CRUD(Create, Read, Update & Delete) operations. Execute MangoDB basic Queries using CRUD operations.	CO5, CO6



DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND DATA SCIENCE

Experiment No 1:

Create a table called Employee & execute the following.

Employee(EMPNO,ENAME,JOB, MANAGER_NO, SAL, COMMISSION)

1. Create a user and grant all permissions to the user.
2. Insert the any three records in the employee table contains attributes

EMPNO,ENAME JOB, MANAGER_NO, SAL, COMMISSION and use rollback. Check the result.

3. Add primary key constraint and not null constraint to the employee table.
4. Insert null values to the employee table and verify the result.

Course outcome:

CO3: Apply various Structured Query Language (SQL) statements for database manipulation.

CO5: Develop database applications for the given real world problem.

Program

Step 1: Create a user and grant all permissions to the user

```
CREATE USER 'new_user'@'localhost' IDENTIFIED BY 'password';
```

```
GRANT ALL PRIVILEGES ON . TO 'new_user'@'localhost';
```

Step 2: Create the Employee table

```
CREATE TABLE Employee (
```

```
EMPNO INT,
```

```
ENAME VARCHAR(50),
```

```
JOB VARCHAR(50),
```

```
MANAGER_NO INT,
```

```
SAL DECIMAL(10,2),
```



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COMMISSION DECIMAL(10,2)

);

Step 3: Insert three records into the Employee table and rollback

START TRANSACTION;

INSERT INTO Employee (EMPNO, ENAME, JOB, MANAGER_NO, SAL,
COMMISSION) VALUES

(1, 'John Doe', 'Manager', NULL, 5000, 1000),

(2, 'Jane Smith', 'Analyst', 1, 4000, 500),

(3, 'Michael Johnson', 'Developer', 1, 4500, NULL);

ROLLBACK;

Check the result (no records should be inserted as the transaction was rolled back)

SELECT * FROM Employee;

Step 4: Add primary key constraint and not null constraint to the employee table

ALTER TABLE Employee

ADD PRIMARY KEY (EMPNO),

MODIFY ENAME VARCHAR(50) NOT NULL,

MODIFY JOB VARCHAR(50) NOT NULL,

MODIFY SAL DECIMAL(10,2) NOT NULL;

Step 5: Insert null values into the Employee table and verify the result

Attempt to insert a record with NULL values



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```
INSERT INTO Employee (EMPNO, ENAME, JOB, MANAGER_NO, SAL,  
COMMISSION) VALUES
```

```
(4, 'Alice', NULL, NULL, NULL, NULL);
```

Check the result (the insertion should fail due to the NOT NULL constraint)

```
SELECT * FROM Employee;
```

OUTPUT



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Experiment No 2:

Create a table called Employee that contain attributes EMPNO,ENAME,JOB, MGR,SAL & execute the following.

1. Add a column commission with domain to the Employee table.
2. Insert any five records into the table.
3. Update the column details of job
4. Rename the column of Employ table using alter command.
5. Delete the employee whose Empno is 105.

CO3: Apply various Structured Query Language (SQL) statements for database manipulation.

CO5: Develop database applications for the given real world problem. **Program**

1. Creating the Employee table with the specified attributes

```
CREATE TABLE Employee (  
    EMPNO INT PRIMARY KEY,  
    ENAME VARCHAR(50),  
    JOB VARCHAR(50),  
    MGR INT,  
    SAL DECIMAL(10, 2)  
);
```

2. Adding a column commission to the Employee table ALTER TABLE Employee
ADD COLUMN COMMISSION DECIMAL(10, 2);

3. Inserting five records into the table

```
INSERT INTO Employee (EMPNO, ENAME, JOB, MGR, SAL, COMMISSION)  
VALUES  
(101, 'John', 'Manager', NULL, 5000.00, 100.00),  
(102, 'Alice', 'Developer', 101, 4000.00, 50.00),
```



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(103, 'Bob', 'Analyst', 101, 3500.00, NULL),
(104, 'Emily', 'Designer', 101, 3800.00, 75.00),
(105, 'Sam', 'Engineer', 101, 4200.00, 80.00);

4. Updating the column details of job

UPDATE Employee

SET JOB = 'Senior ' || JOB

WHERE SAL > 4000;

5. Renaming the column of Employee table using ALTER command

ALTER TABLE Employee

RENAME COLUMN ENAME TO EMP_NAME;

6. Deleting the employee whose Empno is 105

DELETE FROM Employee

WHERE EMPNO = 105;

OUTPUT



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Experiment No 3:

Queries using aggregate functions(COUNT,AVG,MIN,MAX,SUM),Groupby, Orderby.

Employee (E_id, E_name, Age, Salary)

1. Create Employee table containing all Records E_id, E_name, Age, Salary.
2. Count number of employee names from employeetable
3. Find the Maximum age from employee table.
4. Find the Minimum age from employeetable.
5. Find salaries of employee in Ascending Order.
6. Find grouped salaries of employees.

CO3: Apply various Structured Query Language (SQL) statements for database manipulation.

CO5: Develop database applications for the given real world problem.

Program

Create the Employee table containing all Records E_id, E_name, Age, Salary

```
CREATE TABLE Employee (
```

```
E_id INT PRIMARY KEY,
```

```
E_name VARCHAR(255),
```

```
Age INT,
```

```
Salary DECIMAL(10, 2)
```

```
);
```

```
INSERT INTO Employee (E_id, E_name, Age, Salary) VALUES
```

```
(1, 'John Doe', 25, 50000.00),
```

```
(2, 'Jane Smith', 30, 60000.00),
```

```
(3, 'Alice Johnson', 28, 55000.00),
```

```
(4, 'Chris Lee', 35, 75000.00),
```

```
(5, 'Emma Davis', 22, 48000.00);
```

Count number of employee names from Employee table:

```
SELECT COUNT(E_name) AS NumberOfEmployees
```

```
FROM Employee;
```



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Find the Maximum age from Employee table:

```
SELECT MAX(Age) AS MaxAge
```

```
FROM Employee;
```

Find the Minimum age from Employee table:

```
SELECT MIN(Age) AS MinAge
```

```
FROM Employee;
```

Find salaries of employees in Ascending Order:

```
SELECT Salary
```

```
FROM Employee
```

```
ORDER BY Salary ASC;
```

Find grouped salaries of employees:

```
SELECT Salary, COUNT(*) AS NumberOfEmployees
```

```
FROM Employee
```

```
GROUP BY Salary;
```

OUTPUT



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Experiment No 4:

Create a row level trigger for the customers table that would fire for INSERT or UPDATE or DELETE operations performed on the CUSTOMERS table. This trigger will display the salary difference between the old & new Salary. CUSTOMERS (ID,NAME,AGE,ADDRESS,SALARY) **CO3:** Apply various Structured Query Language (SQL) statements for database manipulation.

CO5: Develop database applications for the given real world problem.

Program

```
CREATE TABLE customers (  
    ID INT PRIMARY KEY,  
    NAME VARCHAR(50),  
    AGE INT,  
    ADDRESS VARCHAR(100),  
    SALARY DECIMAL(10, 2)  
);
```

```
CREATE TABLE trigger_log (  
    LOG_ID INT PRIMARY KEY AUTO_INCREMENT,  
    MESSAGE VARCHAR(200)  
);
```

```
DELIMITER //
```

```
CREATE TRIGGER after_customers_insert  
AFTER INSERT ON customers  
FOR EACH ROW  
BEGIN  
    DECLARE salary_diff DECIMAL(10, 2);  
    DECLARE message_text VARCHAR(200);
```



```
SET salary_diff = NEW.SALARY;

SET message_text = CONCAT('Salary difference (new - old) for ID ', NEW.ID, ' is ', salary_diff);

INSERT INTO trigger_log (message) VALUES(message_text); END;

//

DELIMITER ;

DELIMITER //

CREATE TRIGGER after_customers_update
AFTER UPDATE ON customers
FOR EACH ROW
BEGIN
    DECLARE salary_diff DECIMAL(10, 2);
    DECLARE message_text VARCHAR(200);

    IF OLD.SALARY <> NEW.SALARY
    THEN
        SET salary_diff = NEW.SALARY - OLD.SALARY;
        SET message_text = CONCAT('Salary difference (new - old) for ID ', NEW.ID, ' is ', salary_diff);
        INSERT INTO trigger_log (message) VALUES(message_text);
    END IF;
END; //

DELIMITER ;

DELIMITER //
```



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```
CREATE TRIGGER after_customers_delete
AFTER DELETE ON customers
FOR EACH ROW
BEGIN
    DECLARE salary_diff DECIMAL(10, 2);
    DECLARE message_text VARCHAR(200);
    SET salary_diff = OLD.SALARY;
    SET message_text = CONCAT('Salary difference (new - old) for ID ', OLD.ID, ' is ', salary_diff);
    INSERT INTO trigger_log (message) VALUES(message_text); END;
//

DELIMITER ;
```

OUTPUT



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Experiment No 5:

Create cursor for Employee table & extract the values from the table. Declare the variables ,Open the cursor & extract the values from the cursor. Close the cursor. Employee(E_id, E_name, Age, Salary)

CO3: Apply various Structured Query Language (SQL) statements for database manipulation.

CO5: Develop database applications for the given real world problem.

Program

USE COMPANY;

CREATE TABLE Employee2 (E_id INT, E_name VARCHAR (255), Age INT, Salary DECIMAL (10, 2));

INSERT INTO Employee2 (E_id, E_name, Age, Salary)

VALUES

(1, 'Samarth', 30, 50000.00),

(2, 'Ramesh Kumar', 25, 45000.00),

(3, 'Seema Banu', 35, 62000.00),

(4, 'Dennis Anil', 28, 52000.00),

(5, 'Rehman Khan', 32, 58000.00);

DELIMITER //

CREATE PROCEDURE fetch_employee_data ()

BEGIN

-- Declare variables to store cursor values -- DECLARE

emp_id INT;

DECLARE emp_name VARCHAR (255);

DECLARE emp_age INT;

DECLARE emp_salary DECIMAL (10, 2);

-- Declare a cursor for the Employee table--

DECLARE emp_cursor CURSOR FOR

SELECT E_id, E_name, Age, Salary



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FROM Employee2;

-- Declare a continue handler for the cursor--

DECLARE CONTINUE HANDLER FOR NOT FOUND

SET @finished = 1;

-- Open the cursor

OPEN emp_cursor;

-- Initialize a variable to control cursor loop

SET @finished = 0;

-- Loop through the cursor results cursor_loop:

LOOP

-- Fetch the next row from the cursor into variables

FETCH emp_cursor INTO emp_id, emp_name, emp_age, emp_salary;

-- Check if no more rows to fetch

IF @finished = 1 THEN

LEAVE cursor_loop;

END IF;

-- Output or process each row (for demonstration, print the values)

SELECT CONCAT ('Employee ID:', emp_id, 'Name: ', emp_name, ', Age: ', emp_age, ', Salary:

', emp_salary) AS Employee_Info;

END LOOP;

-- Close the cursor

CLOSE emp_cursor;

END//

DELIMITER;



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CALL fetch_employee_data();

OUTPUT



Experiment No 6:

Write a PL/SQL block of code using parameterized Cursor, that will merge the data available in the newly created table N_RollCall with the data available in the table O_RollCall. If the data in the first table already exist in the second table then that data should be skipped.

CO3: Apply various Structured Query Language (SQL) statements for database manipulation.

CO6: Understand the concepts related to NoSQL databases.

Program

```
CREATE DATABASE ROLLCALL;
```

```
USE ROLLCALL;
```

```
-- Create N_RollCall table--
```

```
CREATE TABLE N_RollCall ( student_id
```

```
INT PRIMARY KEY, student_name
```

```
VARCHAR(255), birth_date DATE
```

```
);
```

```
-- Create O_RollCall table with common data
```

```
CREATE TABLE O_RollCall ( student_id
```

```
INT PRIMARY KEY, student_name
```

```
VARCHAR(255), birth_date DATE
```

```
);
```

```
-- Insert common data into O_RollCall--
```

```
INSERT INTO O_RollCall (student_id, student_name, birth_date)
```

```
VALUES
```

```
(1, 'Shivanna', '1995-08-15');
```



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(3, 'Cheluva', '1990-12-10');

-- Insert sample records into N_RollCall--

INSERT INTO N_RollCall (student_id, student_name, birth_date)

VALUES

(1, 'Shivanna', '1995-08-15'); -- Common record with O_RollCall

(2, 'Bhadramma', '1998-03-22');

(3, 'Cheluva', '1990-12-10'); -- Common record with O_RollCall

(4, 'Devendra', '2000-05-18');

(5, 'Eshwar', '1997-09-03');

-- merge_rollcall_data stored procedure---

DELIMITER //

CREATE PROCEDURE merge_rollcall_data()

BEGIN

DECLARE done INT DEFAULT FALSE;

DECLARE n_id INT;

DECLARE n_name VARCHAR(255);

DECLARE n_birth_date DATE;

-- Declare cursor for N_RollCall table--

DECLARE n_cursor CURSOR FOR

SELECT student_id, student_name, birth_date

FROM N_RollCall;

-- Declare handler for cursor--

DECLARE CONTINUE HANDLER FOR NOT FOUND

SET done = TRUE;



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-- Open the cursor--

OPEN n_cursor;

-- Start looping through cursor results

cursor_loop: LOOP

-- Fetch data from cursor into variables

FETCH n_cursor INTO n_id, n_name, n_birth_date;

-- Check if no more rows to fetch

IF done THEN

LEAVE cursor_loop;

END IF;

-- Check if the data already exists in O_RollCall

IF NOT EXISTS (

SELECT 1

FROM O_RollCall

WHERE student_id = n_id

) THEN

-- Insert the record into O_RollCall

INSERT INTO O_RollCall (student_id, student_name, birth_date)

VALUES (n_id, n_name, n_birth_date);

END IF;

END LOOP;

CLOSE n_cursor;

END//



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DELIMITER ;

--execute the merge_rollcall_data stored procedure

CALL merge_rollcall_data(); --

Select all records from O_RollCall

SELECT * FROM O_RollCall;

OUTPUT



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Experiment No 7:

Install an Open Source NoSQL Data base MangoDB& perform basic CRUD(Create, Read, Update & Delete) operations. Execute MangoDB basic Queries using CRUD operations.

```
[
  {
    "id":123,
    "name":"KGF 1",
    "theme":"Action"
  },
  {
    "id":234,
    "name":"KGF 2",
    "theme":"Action"
  },
  {
    "id":456,
    "name":"Bahubali 1",
    "theme":"Fiction"
  },
  {
    "id":789,
    "name":"Bahubali 2",
```



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"theme": "Fiction"

},

{

"id": 123,

"name": "RRR",

"theme": "Historical"

}

]

Open MongoDB and import data

Test > use hobbies,

<switch to db hobbies,

>bd. movie. insertOne ({ "id" : 395 , "name": "3 idiots", "theme": "comedy" })

<{

Acknowledged : true, insertedId :

ObjectId('6694bba1ba2751895234ab45')

}

>bd. movie. find ({ "id" : 123 , "name": "KGF 1", "theme": "Action" })

<{

_id: ObjectId('6694bba1ba2751895234ab45')

Id: 123,

Name : 'KGF 1'



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Theme : 'Action'

}

```
>db.movie.deleteOne({"id":456 , "name" : "Bahubali 1" , "theme" : "Fiction"}) {
```

Acknowledged : true,

Deleted count : 1

```
>db.movie.updateOne : ({"theme" : "comedy"},{$ set : { " theme" : "funny" } }) {
```

Acknowledged : true, inserted : null,

matchedCount : 1,

modifiedCount : 1,

upsertedCount : 0

}

OUTPUT



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INSTITUTION VISION & MISSION

Vision

"To become a recognized world class Women Educational Institution, by imparting professional education to the students, creating technical opportunities through academic excellence and technical achievements, with ethical values"

Mission

- To support value based education with state of art infrastructure.
- To empower women with the additional skill for professional future career.
- To enrich students with research blends in order to fulfill the International Challenges.
- To create multidisciplinary center of excellence.
- To achieve Accreditation standards towards International education recognition.
- To establish more Post Graduate & Research course.
- To increase Doctorates numbers towards the Research quality of academics.